

Milestone 0  
By Anthony and Ivy

## Dataset

Our dataset is titled “[PM25Vision: Global images with PM2.5 labels](#)” with the whole dataset for machine learning linked [here](#). This dataset has a bunch of images taken from a variety of cameras, all with the sky present in part of the picture. Using images as input, the model is meant to identify the sky and output the PM2.5 concentration that it predicts from these images. Then, the model will classify these concentrations into different class levels, from 0 (*most clear non-toxic*) to class 5 (*hazardous*). This dataset has 11,219 datapoints and has already been split between training and testing data. At the bottom of the dataset, there are sample codes that we can use to classify a range of outputs into class level. The dataset also mentions the underrepresentative of the extremes, which might have us scale the dataset to receive more varied predictions.

## Model Architecture

| Air Quality Index                        |                |                                                                                                                                           |
|------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| AQI Category and Color                   | Index Value    | Description of Air Quality                                                                                                                |
| Good<br>Green                            | 0 to 50        | Air quality is satisfactory, and air pollution poses little or no risk.                                                                   |
| Moderate<br>Yellow                       | 51 to 100      | Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution. |
| Unhealthy for Sensitive Groups<br>Orange | 101 to 150     | Members of sensitive groups may experience health effects. The general public is less likely to be affected.                              |
| Unhealthy<br>Red                         | 151 to 200     | Some members of the general public may experience health effects; members of sensitive groups may experience more serious health effects. |
| Very Unhealthy<br>Purple                 | 201 to 300     | Health alert: The risk of health effects is increased for everyone.                                                                       |
| Hazardous<br>Maroon                      | 301 and higher | Health warning of emergency conditions: everyone is more likely to be affected.                                                           |

Since we have chosen an output that is a bit more abstract than a typical classification, we are following a *CNN structure* and going to feed an image to predict a float. Then, round that value to the nearest integer. Finally have that int be associated with a class for Air Quality Index (AQI) classification 0 (safest) - 5 (hazardous). We are using the standard AQI table to help identify ranges for classes.

#### **Ranges:**

**0-50** - class 0

**51 - 100** - class 1

**101 - 150** - class 2

**151 - 200** - class 3

**201 - 300** - class 4

**301 +** - class 5

## Accuracy Prediction

We expect an accuracy of 30% with more or less around there. AQI levels are abstract for human eyes, and similarly for models! Our model has a decent amount of samples, but the portion of the sky/quality of the street-level imagery varies a good amount which could make it hard for the model to identify. Along with that, the amount of possible predictions for concentration also could make the model have a lower accuracy potential due to rounding differences, since a prediction of 50.1 is close to but not the same as 51.1 and might be satisfactory for other prediction cases; however, considered different enough that AQI classes place them at different classes Good(1) & Moderate(1). The dataset also contains an underrepresented amount of extreme high/low PM2.5 images, which can skew the results.

## Weekly Meeting Time/Date

Anthony has SJCC classes until next week, but I have an event to attend the week after. We decided to meet Wednesday 8pm-10pm in discord and for the last week, meet up right before class on Thursday, 11am-1pm.

Week 8+9: Wed 8pm-10pm (Discord Call)

Week 10: Thurs 11am-1pm (Baskin Engineering in person)

## Timeline/Work Plan

#### **General:**

We plan to discuss planning, catchup, and what needs to get done for the next week at our meetings on Wednesday. On Friday, we hope to have more work done, so we can run over it with

Eric. Anthony plans to attend extra office hours that Ivy couldn't, with Kimbrell, either on Saturday or Wednesday, to ask any questions and will keep me updated.

### **Week 8: Milestone 1**

For the Data Processing step, we will be using a dataloader, similar to the midterm. We will augment the first half, where the format is mostly correct for datasets, with torch.transforms to go with string -> pillow image -> tensor. By this step we should have created the dataset class and have passed it through the dataset loader to see if it works.

Ivy will be in charge of writing headers on the main code signifying where the data processing, data splitting, training, testing, and classes will be placed. Along with that, she will be in charge of loading the data into vscode, splitting, and cleaning to be readied for usage. We have discussed dropping nulls and duplicates, not yet dropping outliers unless it's really messing up results, and having the outputs be PM2.5.

Anthony would be in charge of creating the Dataloader and implementing image transforms using PIL and Torch's Transform library. Anthony will also implement the batches, enabling shuffling and having batches of 32, for further training and testing.

We will work together on building the block diagram for the project.

### **Week 9: Milestone 2**

By Milestone Model Draft, Ivy will be in charge of creating the Model class and training loop which utilizes the training data and validation data. Anthony will work on adding augmentation to the rest of the dataset, implementing transform.flip/rotate for more image variability. Anthony will also create the Testing loop using Adam Optimization and MSE loss, but consider playing with different layers and activations. We will help each other in parts when needed.

### **Week 10: Final Submission**

By final submission, Ivy and Anthony will be reviewing and debugging, using slides and office hours as resources. Potentially, if there is time, they plan to graph and analyze our models' stats{Loss(y) vs Epoch(x), Accuracy, Precision}. Additionally, we will try to practice going through the presentation and making sure the code works.

### **Github Link:**

<https://github.com/ivjoli/CMPM17-Final-Project-Image-Sensing>